THE SEGAL CONDITION FOR ALL n, COMPOSITION, AND FULLY FAITHFUL FUNCTORS FROM THE SATTLER–WÄRN AXIOMS, AND THEIR FORMALISATION IN UNIT

Abstract. We work in univalent type theory with an axiomatised wild category Cat of higher categories, following Wärn's notes and the axioms of Sattler and Wärn. From the Segal axiom and the fact that Δ^{n+1} is obtained from Δ^n by gluing an edge (Lemma 26 of Wärn's notes) we prove the Segal condition for every n: a functor $\Delta^n \to C$ is the same as a chain of n composable morphisms of C. Composition of morphisms is defined as the centre of a contractible type of fillers, and the unit and associativity laws follow from the cases $n = 2$ and $n = 3$. We then prove Wärn's results 33–39 on fully faithful functors: they reflect invertibility, are embeddings on objects, and are isomorphisms when they are surjective on objects. All of this uses only two coherences of Cat beyond its associativity and unit laws: the pentagon and the triangle. The last section explains how the proofs are formalised in the proof checker Unit.

1. Introduction

Wärn's notes *Higher-categorical headcanon* and the axioms of Sattler and Wärn present $(\infty, 1)$-category theory on top of homotopy type theory, after Cisinski, Cnossen, Nguyen and Walde. One postulates a wild category Cat, whose objects are called categories and whose morphisms are called functors, with a few universal properties: a terminal category, pullbacks, exponentials, the walking arrow $\mathbb{I}$, and the Segal axiom. Everything internal to a category C (its objects, morphisms, composition and the laws of composition) is then constructed. Being a wild category is a matter of coherences: the associativity of functor composition holds up to a path, these paths satisfy the pentagon up to a 2-path, and so on. Wärn treats the whole tower as an axiom schema and expects that every concrete argument uses only finitely many levels.

This note proves the following.

(1) The Segal condition for every n (Theorem 4.3). The case $n = 2$ is an axiom; the other cases follow from it and from Wärn's Lemma 26.
(2) Composition of morphisms with its unit and associativity laws, and the fact that functors preserve composition and identities (Section 5).
(3) Wärn's results 33–39 on invertible morphisms and fully faithful functors (Section 6), with the converse of Remark 39.

We also record which coherences each result uses (Section 7). Composition itself uses only the associativity and unit paths of Cat. The pentagon and the triangle enter because a functor acts on a morphism together with the identifications of its endpoints, and the objects of a category do not form a set. No higher coherence is used.

2. The axioms

We work in dependent type theory with univalence; function extensionality follows. Equality means the identity type, and $p \cdot q$ is the composite of paths.

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2.1. **Level 1: the wild category.** We are given a type Cat of *categories*, for $A, B : \mathsf{Cat}$ a type $\mathrm{Map}(A, B)$ of *functors*, composition $g \circ f$, identities 1_A, and paths

$$\alpha_{h,g,f} : (h \circ g) \circ f = h \circ (g \circ f), \qquad \lambda_f : 1 \circ f = f, \qquad \rho_f : f \circ 1 = f.$$

The types $\mathrm{Map}(A, B)$ are not assumed to be sets. A functor $f : A \to B$ is an *isomorphism* if $f \circ - : \mathrm{Map}(X, A) \to \mathrm{Map}(X, B)$ is an equivalence for every X. This is a proposition, and an isomorphism has a two-sided inverse.

2.2. **Level 2: pentagon and triangle.** We are given the pentagon, a path between the two ways of reassociating $((f \circ g) \circ h) \circ e$ into $f \circ (g \circ (h \circ e))$, and the triangle relating $\alpha_{g,1,f}$, ρ_g and λ_f. The other two triangles and Mac Lane's coherence theorem (any two parallel composites of α, λ, ρ and their inverses are equal) follow, as usual.

2.3. **Structure.** The following axioms only mention level-1 data.

- Cat is univalent (Axiom 3), and has a terminal category $\mathbf{1}$ (Axiom 4), pullbacks (Axiom 13) and exponentials $\mathrm{Fun}(A, B)$ with $\mathrm{Map}(X, \mathrm{Fun}(A, B)) \simeq \mathrm{Map}(X \times A, B)$ (Axiom 15).
- The objects of C are $\mathrm{Ob}C = \mathrm{Map}(\mathbf{1}, C)$, and a functor acts on them by composition.
- There is a category $\mathbb{I}$ with $\mathrm{Ob}\mathbb{I} \simeq \{0, 1\}$ (Axiom 8). Writing $\mathrm{Mor}C = \mathrm{Map}(\mathbb{I}, C)$, $\mathrm{dom}m = m \circ 0$ and $\mathrm{cod}m = m \circ 1$, every morphism of $\mathbb{I}$ goes from its domain to an object that is at least as large. Moreover $\mathrm{Map}(C, \mathbb{I})$ is equivalent to the type of maps $\mathrm{Ob}C \to \{0, 1\}$ that are monotone along every morphism (Axiom 18).
- The Segal axiom (Axiom 21): with $\Delta^0 = \mathbf{1}$ and $\Delta^{n+1} = \mathrm{Fun}(\Delta^n, \mathbb{I})$ (Definition 19), the map

  $$\mathrm{Map}(\Delta^2, C) \to (f, g : \mathrm{Mor}C) \times (\mathrm{cod}f = \mathrm{dom}g), \qquad s \mapsto (s \circ d_{01},\ s \circ d_{12})$$

  is an equivalence. Here d_{01}, d_{12} are the two spine edges of Δ^2.

The *mapping space* is

$$\mathrm{Hom}_C(x, y) = (m : \mathrm{Mor}C) \times (\mathrm{dom}m = x) \times (\mathrm{cod}m = y).$$

Since $\mathrm{Ob}C$ is not a set, a path in $\mathrm{Hom}_C(x, y)$ is a path $m = m'$ of morphisms together with two 2-cells relating the endpoint proofs. The identity of c is $\mathrm{id}_c = c \circ !$ with $! : \mathbb{I} \to \mathbf{1}$, and its endpoint proofs use that $\mathrm{Map}(\mathbf{1}, \mathbf{1})$ is contractible. A functor $F : A \to B$ acts on Hom:

$$F(m, p, q) = (F \circ m,\ \alpha_{F,m,0} \cdot \mathrm{ap}_F\, p,\ \alpha_{F,m,1} \cdot \mathrm{ap}_F\, q) : \mathrm{Hom}_B(Fx, Fy).$$

Two more axioms concern invertible morphisms; they are stated in Section 6, once composition has been defined.

2.4. **Simplices.** From Axiom 18 one gets $\mathrm{Ob}\Delta^n \simeq [n] = \{0 < 1 < \cdots < n\}$, and functors into Δ^n are monotone maps on objects (Construction 20). Every Δ^n and every $\mathrm{Fun}(X, \mathbb{I})$ is *posetal*: its mapping spaces are propositions, its objects form a set, and a functor into it is determined by its action on objects. Consequently $\mathrm{Map}(A, \Delta^n)$ is a set. We write top_n for the last vertex of Δ^n, $\phi_n : \Delta^n \to \Delta^{n+1}$ for the face that omits the last vertex, and $\ell_n : \mathrm{Hom}_{\Delta^{n+1}}(\phi_n(\mathrm{top}_n), \mathrm{top}_{n+1})$ for the last edge.

3. Gluing an edge to a simplex

The *gluing square* S_n is

$$\begin{array}{ccc} \mathbf{1} & \xrightarrow{0} & \Delta^1 \\ {\scriptstyle \mathrm{top}_n}\downarrow & & \downarrow{\scriptstyle \ell_n} \\ \Delta^n & \xrightarrow[\phi_n]{} & \Delta^{n+1} \end{array}$$

where the right edge is ℓ_n precomposed with the isomorphism $\Delta^1 \simeq \mathbb{I}$. A commuting square is a *pushout* when, for every X, maps out of its corner Δ^{n+1} correspond, by precomposition, to cones over its span.

Theorem 3.1 (Wärn, Lemma 26)**.** *For every n the square S_n is a pushout.*

Proof. For $n = 0$ the left edge is an identity. For $n = 1$ the cones over the span are composable pairs, through $\Delta^1 \simeq \mathbb{I}$, and the statement is the Segal axiom. For the induction step we use three facts.

(1) $- \times \mathbb{I}$ preserves pushouts, since it has the right adjoint $\mathrm{Fun}(\mathbb{I}, -)$. So $S_{n+1} \times \mathbb{I}$ is a pushout.
(2) S_{n+2} is a retract of $S_{n+1} \times \mathbb{I}$. At the corners Δ^{k+1} the section is $\psi \mapsto (\psi \circ \phi_k,\ \psi(\mathrm{top}_{k+1}))$, which restricts a functor $\psi : \Delta^{k+1} \to \mathbb{I}$ to the face and evaluates it at the top. The retraction is induced by a monotone retraction of posets $[k+1] \times \{0,1\} \to [k+2]$. At the corners $\mathbf{1}$ and Δ^1 the section is $x \mapsto (x, 1)$ and the retraction is the first projection. That the sides commute is checked on objects, as all the corners are posetal.
(3) A retract of a pushout square is a pushout square, provided the types of functors between corners are sets. Here they are, since the corners are posetal. Under that hypothesis a morphism of squares needs no coherence between its faces, and the comparison map of the retract is a retract of an equivalence. □

The same holds with $\mathbb{I}$ in place of Δ^1. That square is a retract of S_n through the isomorphism $\Delta^1 \simeq \mathbb{I}$. So for every n and C the map

$$\mathrm{Map}(\Delta^{n+1}, C) \to (a : \mathrm{Map}(\Delta^n, C)) \times (m : \mathrm{Mor}C) \times \big(a(\mathrm{top}_n) = \mathrm{dom}m\big), \tag{3.1}$$

which sends b to $b \circ \phi_n$ and $b \circ \ell_n$ together with the path given by the commuting square, is an equivalence.

4. The Segal condition for all n

Definition 4.1. The type $\mathrm{Ch}_n(x)$ of *chains* of n composable morphisms ending at $x : \mathrm{Ob}C$ is defined by recursion on n:

$$\mathrm{Ch}_0(x) = \mathbf{1}, \qquad \mathrm{Ch}_{n+1}(x) = (y : \mathrm{Ob}C) \times \mathrm{Ch}_n(y) \times \mathrm{Hom}_C(y, x).$$

So a chain of length $n+1$ is a chain of length n with a morphism glued at its last vertex. A functor F maps chains to chains, vertex by vertex and edge by edge. The *universal chain* $u_n : \mathrm{Ch}_n(\mathrm{top}_n)$ of Δ^n is defined by $u_0 = \star$ and $u_{n+1} = (\phi_n(\mathrm{top}_n),\ \phi_n(u_n),\ \ell_n)$. The *spine* of $s : \Delta^n \to C$ is $s(u_n) : \mathrm{Ch}_n(s(\mathrm{top}_n))$.

Lemma 4.2. *For $G : A \to B$, $F : B \to C$ and $h : \mathrm{Hom}_A(x, y)$, the morphism $(F \circ G)(h)$, with its endpoints moved along α, is equal to $F(G(h))$. Consequently $(F \circ G)(c) = F(G(c))$ for chains c, over α at the last vertex. Also $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$, and $m : \mathrm{Mor}C$ is the image of the generating arrow of $\mathbb{I}$.*

Proof. By path induction on the endpoint proofs of h, we may take $h = (m, \mathrm{refl}, \mathrm{refl})$. The morphisms agree by $\alpha_{F,G,m}$. The two endpoint 2-cells are the pentagon at $(F, G, m, 0)$ and at $(F, G, m, 1)$. For chains, induct on the length. For the identity, the morphisms agree by $\alpha^{-1}_{F,c,!}$, and the endpoint 2-cells agree by the pentagon at $(F, c, !, 0)$, naturality of α, and the triangle. The last statement is the triangle at $(0, m)$ and $(1, m)$. □

Theorem 4.3 (Segal condition)**.** *For every n and every category C, the spine map*

$$\mathrm{Map}(\Delta^n, C) \longrightarrow (x : \mathrm{Ob}C) \times \mathrm{Ch}_n(x), \qquad s \mapsto (s(\mathrm{top}_n),\ s(u_n))$$

is an equivalence.

Proof. By induction on n. For $n = 0$, both sides are ObC. For $n+1$, the spine map is homotopic to the composite of three equivalences:

$$\mathrm{Map}(\Delta^{n+1}, C) \;\simeq\; (a, m, p) \;\simeq\; ((x, c), m,\, x = \mathrm{dom} m) \;\simeq\; (x : \mathrm{Ob}C) \times \mathrm{Ch}_{n+1}(x).$$

The first is (3.1). The second applies the induction hypothesis to a. The third glues: $((x, c), m, p) \mapsto (\mathrm{cod} m,\, x,\, c,\, (m, p^{-1}, \mathrm{refl}))$, an equivalence by contracting the singleton $(\mathrm{cod} m, \mathrm{refl})$. To see that the composite is the spine map, take $b : \Delta^{n+1} \to C$.

- The last vertices agree, since $\mathrm{cod}(b \circ \ell_n) = b(\mathrm{top}_{n+1})$.
- The chains agree: the spine of $b \circ \phi_n$ is $(b \circ \phi_n)(u_n) = b(\phi_n(u_n))$ by Lemma 4.2.
- The last morphisms agree by a reassociation of the commuting square. □

Corollary 4.4. *The type of n-simplices of C with a given spine is contractible.*

5. Composition and its laws

Let Pair_C be the type of composable pairs with their vertices, (z, y, x) with $f : \mathrm{Hom}(z, y)$ and $g : \mathrm{Hom}(y, x)$. The *boundary* of $s : \Delta^2 \to C$ is the image under s of the three edges d_{01}, d_{12}, d_{02} of Δ^2 with their endpoints. Its first two edges form a pair $\partial s : \mathrm{Pair}_C$.

Lemma 5.1. *$\partial : \mathrm{Map}(\Delta^2, C) \to \mathrm{Pair}_C$ is an equivalence. Its fibres, the* fillers *of a pair, are contractible.*

Proof. ∂ is the Segal comparison followed by the equivalence $(f, g, p) \mapsto (\mathrm{dom} f, \mathrm{cod} f, \mathrm{cod} g, (f, \mathrm{refl}, \mathrm{refl}), (g, p^{-1}, \mathrm{refl}))$. The inverse of that equivalence forgets the vertices, which works by contracting the endpoint proofs. The two constructions agree on the path $\mathrm{cod}(s\, d_{01}) = \mathrm{dom}(s\, d_{12})$ by the groupoid laws. □

Lemma 5.1 uses only the Segal axiom, not Theorem 4.3. The spine of a 2-simplex, read as a pair, is exactly ∂s. This is because the spine of Δ^2 consists of the edges d_{01} and d_{12}, and pairs in the posetal Δ^2 are determined by their vertices. So Lemma 5.1 is the case $n = 2$ of Theorem 4.3.

Definition 5.2 (Construction 22)**.** For a pair (f, g), the composite $g \circ f : \mathrm{Hom}(z, x)$ is the third edge of the centre of its type of fillers, transported along the identification of the centre's first two edges with (f, g).

Composition is characterised by two properties.

(1) If $s : \Delta^2 \to C$ has boundary (f, g, h), then $h = g \circ f$. This holds because $(s, \ldots)$ is a filler and fillers form a contractible type.
(2) Every pair (f, g) has a 2-simplex with boundary $(f, g, g \circ f)$.

In particular, on the boundary of any 2-simplex, $g \circ f$ has the same underlying morphism as the long edge of the witness given by the Segal axiom.

Proposition 5.3. *For a functor $F : A \to C$: $F(g \circ f) = Fg \circ Ff$ and $F(\mathrm{id}_x) = \mathrm{id}_{Fx}$.*

Proof. Let s have boundary $(f, g, g \circ f)$. By Lemma 4.2 applied to each edge, $F \circ s$ has boundary $(Ff, Fg, F(g \circ f))$. Property (1) of composition gives the first statement. The second is in Lemma 4.2. □

Proposition 5.4 (Construction 23)**.** *$h \circ \mathrm{id}_a = h$ and $\mathrm{id}_b \circ h = h$ for $h : \mathrm{Hom}_C(a, b)$.*

Proof. By path induction on the endpoint proofs, take $h = (m, \mathrm{refl}, \mathrm{refl})$ with $m : \mathbb{I} \to C$. In $\mathbb{I}$ the pair (id_0, γ), where γ is the generating arrow, has a filler σ by Lemma 5.1. Its long edge is γ, since $\mathrm{Hom}_{\mathbb{I}}(0, 1)$ is a proposition. By Lemma 4.2, $m \circ \sigma$ has boundary $(m\, \mathrm{id}_0,\, m\gamma,\, m\gamma) = (\mathrm{id}_a, h, h)$. So $h \circ \mathrm{id}_a = h$ by property (1). The other law uses the pair (γ, id_1). □

Proposition 5.5 (Construction 27)**.** *$(k \circ g) \circ f = k \circ (g \circ f)$ for composable f, g, k.*

Proof. By Theorem 4.3 at $n = 3$, the type of triples of composable morphisms with their vertices is equivalent to $\mathrm{Map}(\Delta^3, C)$, through the spine. It therefore suffices to prove the law for the spine of some $T : \Delta^3 \to C$, that is, for $(Tk_{01}, Tk_{12}, Tk_{23})$ where $k_{i,i+1}$ is the spine of Δ^3. By Proposition 5.3 both sides are images under T:

$$(Tk_{23} \circ Tk_{12}) \circ Tk_{01} = T((k_{23} \circ k_{12}) \circ k_{01}), \qquad Tk_{23} \circ (Tk_{12} \circ Tk_{01}) = T(k_{23} \circ (k_{12} \circ k_{01})).$$

The two composites in Δ^3 are equal, because Δ^3 is posetal and its mapping spaces are propositions. □

6. Invertible morphisms and fully faithful functors

Definition 6.1 (Definitions 33 and 34)**.** A morphism $f : \mathrm{Hom}_C(a, b)$ is *invertible* if, for every x, postcomposition $f \circ - : \mathrm{Hom}(x, a) \to \mathrm{Hom}(x, b)$ is an equivalence. This is a proposition. Let $\mathrm{Iso}_C(a, b)$ be the type of invertible morphisms. A functor F is *fully faithful* if each action $\mathrm{Hom}_A(a_0, a_1) \to \mathrm{Hom}_B(Fa_0, Fa_1)$ is an equivalence.

Proposition 6.2. *f is invertible if and only if it has a two-sided inverse. Identities are invertible, and functors preserve invertibility.*

Proof. If g is a two-sided inverse, then $g \circ -$ is inverse to $f \circ -$, by associativity and the unit laws. Conversely, let g be the preimage of id_b under $f \circ -$. Then $f \circ g = \mathrm{id}_b$. Moreover $f \circ (g \circ f) = (f \circ g) \circ f = f = f \circ \mathrm{id}_a$, and $f \circ -$ is injective, so $g \circ f = \mathrm{id}_a$. Identities have themselves as inverses. Functors preserve two-sided inverses by Proposition 5.3. □

Lemma 6.3 (Lemma 35)**.** *A fully faithful functor reflects invertibility.*

Proof. By Proposition 5.3, $F \circ (f \circ -) = (Ff \circ -) \circ F$ as maps $\mathrm{Hom}(x, a) \to \mathrm{Hom}(Fx, Fb)$. If Ff is invertible and F is fully faithful, the right side is an equivalence. So $f \circ -$ is one by cancellation. □

Axiom 6.4 (Rezk, Axiom 36)**.** For every $a : \mathrm{Ob}C$, the type $(b : \mathrm{Ob}C) \times \mathrm{Iso}_C(a, b)$ is contractible.

Remark 6.5. Wärn states Axiom 36 as follows: $c \mapsto \mathrm{id}_c$, $\mathrm{Ob}C \to \mathrm{Mor}C$, is an embedding whose image contains the invertible morphisms. The two forms are equivalent. Since $\mathrm{Mor}C \simeq (a, b : \mathrm{Ob}C) \times \mathrm{Hom}(a, b)$ by contracting the endpoint proofs, and invertibility is a proposition, Wärn's form says that $\iota : c \mapsto (c, c, \mathrm{id}_c)$ is an equivalence $\mathrm{Ob}C \simeq (a : \mathrm{Ob}C) \times (b : \mathrm{Ob}C) \times \mathrm{Iso}(a, b)$. This holds because a surjective embedding is an equivalence, and identities are invertible. As ι lies over the identity of $\mathrm{Ob}C$, it is an equivalence exactly when each map $\mathbf{1} \to (b : \mathrm{Ob}C) \times \mathrm{Iso}(a, b)$ is one, which is Axiom 6.4.

By the fundamental theorem of identity types, Axiom 6.4 is equivalent to the following: the map $\mathrm{idtoiso} : (a = b) \to \mathrm{Iso}(a, b)$, which transports id_a, is an equivalence.

Lemma 6.6 (Lemma 37)**.** *If F is fully faithful, then $\mathrm{Ob}F : \mathrm{Ob}A \to \mathrm{Ob}B$ is an embedding.*

Proof. The action of F on invertible morphisms, $\mathrm{Iso}(a_0, a_1) \to \mathrm{Iso}(Fa_0, Fa_1)$, is an equivalence. It is the action on Hom, an equivalence, over a map between propositions that goes both ways by Proposition 6.2 and Lemma 35. By path induction and $F(\mathrm{id}) = \mathrm{id}$, we have $\mathrm{idtoiso}(\mathrm{ap}_F\, p) = F(\mathrm{idtoiso}\, p)$. So $\mathrm{ap}_F = \mathrm{idtoiso}^{-1} \circ F \circ \mathrm{idtoiso}$ is an equivalence. □

Axiom 6.7 ($\mathbb{I}$ detects equivalences, Axiom 38)**.** If $F \circ - : \mathrm{Map}(\mathbb{I}, A) \to \mathrm{Map}(\mathbb{I}, B)$ is an equivalence, then F is an isomorphism.

Theorem 6.8 (Remark 39 and its converse)**.** *A functor $F : A \to B$ is an isomorphism if and only if it is fully faithful and surjective on objects. Here surjective means that every fibre of* $\mathrm{Ob}F$ *is merely inhabited.*

Proof. Suppose F is fully faithful and surjective. By Lemma 37, ObF is an embedding. Being surjective, it is an equivalence. Through Mor$C \simeq (a, b) \times \mathrm{Hom}(a, b)$, postcomposition $F \circ -$ on morphisms becomes $((a, b), h) \mapsto ((Fa, Fb), Fh)$. This is a fibrewise equivalence followed by a change of base along the equivalence Ob$F \times$ ObF, so it is an equivalence. Axiom 38 then shows that F is an isomorphism.

Conversely, suppose F is an isomorphism. Postcomposition at $X = \mathbf{1}$ shows that ObF is an equivalence, hence surjective. Postcomposition at $X = \mathbb{I}$ shows that the total map above is an equivalence. After cancelling the change of base, the map over Ob$A \times$ ObA with fibres $h \mapsto Fh$ is an equivalence of total spaces. So it is a fibrewise equivalence, which says that F is fully faithful. □

7. Which coherences are used

Result	Level	Coherence used
objects, morphisms, Δ^n, Construction 20, posetality	1	–
$\mathbb{I}$-faces of Δ^2, retract $S_{n+2} < S_{n+1} \times \mathbb{I}$	1	–
composition, its characterisation (Lemma 5.1)	1	–
definitions of invertible, fully faithful; Axiom 36	1	–
Lemma 4.2	2	pentagon, triangle
Theorem 3.1 (Lemma 26)	2	pentagon (cones, products)
Theorem 4.3 (Segal for all n)	2	Lemma 26, Lemma 4.2
unit laws, associativity, Proposition 5.3	2	Lemma 4.2
results 35–39	2	through the above

Nothing uses a coherence of level 3. Level 2 enters in two places:

- in the algebra of cones and products used for Lemma 26;
- in the action of functors on morphisms together with their endpoint identifications (Lemma 4.2).

The second is needed because ObC is not a set.

8. The formalisation

Everything above is checked in Unit, a proof checker for dependent type theory (`github.com/coquand/unit`), in the directory `unit/sag/hcat`. Sections 2–3 port an Agda proof of Lemma 26 (`github.com/coquand/basic-higher-category`). Sections 4–7 are new.

8.1. **Axioms as nested segments.** Unit has *segments*: modules whose parameters are assumptions, and which can be nested. The root file `All.seg` is laid out as

`Theory` ⊃ `Wild` ⊃ `Structure` ⊃ `Coh2`.

- `Theory` takes the parameters `univ` (univalence), `images` and `surjections` (replacement). Function extensionality is derived.
- `Wild` takes the level-1 data of `Cat`.
- `Structure` takes the axioms of Section 2 together with `rezk` and `I-detects-equiv`.
- `Coh2` takes `pentagonator` and `triangle2`.

Each file is imported in the outermost segment whose parameters it uses, so the level column of Section 7 can be read from the layout. The definition of `Cat` has no judgemental shortcuts: the laws of functor composition are paths, and the two triangles that are not parameters, as well as Mac Lane coherence, are proved from the pentagon and `triangle2`. The formalisation uses no

new axioms beyond the `Theory` parameters `univ`, `images`, `surjections` and the parameters of `Wild`, `Structure` and `Coh2`.

8.2. **Where the results are.**

Result	File and name
Theorem 3.1	`Lemma26.lemma26-square`
the square with $\mathbb{I}$, (3.1)	`SpineSquare.pushout-is-equiv`
chains, u_n, spine	`Chains.HChain`, `univ-chain`, `spine-total`
Lemma 4.2	`HomFunctor.Hom-map-of-comp`, `Hom-map-id`, `Hom-map-gen`; `ChainFunctor.Chain-map-of-comp`
Theorem 4.3, corollary	`SegalAll.segal-all`, `filler-contr`
Lemma 5.1, composition	`Composition.segal2-equiv`, `composite` (locked), `composite-witness`, `composite-filler`, `hcomp`
spine at $n = 2$ is ∂	`SpineTwo.spine-two`
agreement with the Segal witness	`Composition.composite-agrees`
Proposition 5.3, unit laws	`CompLaws.Hom-map-hcomp`, `hcomp-unit-r`, `hcomp-unit-l`
associativity	`Associativity.hcomp-assoc`
invertible, fully faithful	`Invertible.is-inv`, `is-inv-prop`, `ObIso`, `is-ff`
Proposition 6.2	`FullyFaithful.has-inv-is-inv`, `is-inv-has-inv`, `id-is-inv`, `functor-pres-inv`
Lemma 35	`FullyFaithful.ff-reflects-inv`
idtoiso is an equivalence	`FullyFaithful.rezk-idtoiso`
Lemma 37	`FullyFaithful.ff-emb`
Remark 39 and converse	`FullyFaithful.ff-surj-iso`, `iso-ff`, `iso-surj`

Composition is written in diagrammatic order in the files: `hcomp g f` is $f \circ g$.

8.3. **Correspondence and differences.**

- Each file begins with the informal proof given here, in more detail. A construction is locked (made opaque) right after the lemmas that characterise it. For example, `composite` is locked after properties (1) and (2) of Section 5, and every later proof uses only those two properties.
- The Agda proof postulates three facts: functors preserve composition, identities are invertible, and invertibility is a proposition. Here they are theorems (Proposition 5.3, Proposition 6.2, and the definition). The three postulates of the Agda proof of Lemma 26 that were candidates for proof are proved as well: $\mathrm{Hom}_{\mathbb{I}}(0, 1)$ is contractible, and the spine edges of Δ^2 are its geometric faces.
- Axiom 6.4 is the parameter `rezk`. The equivalence with Wärn's formulation (Remark 6.5) is proved here on paper but not formalised. Its ingredients are in the files (`mor-ends-equiv`, `is-inv-prop`, `fibrewise-from-total`).
- The agreement of composition with the Segal witness is formalised for the underlying morphisms only, not for their endpoint proofs.
- Surjectivity is propositionally truncated, using the truncation of the foundations.

8.4. **Figures.** Checking the whole directory takes 88 modules and 1674 declarations, with no failures, in about 7 s. Each file of Sections 4–6 checks in under 0.2 s.

8.5. **Open.** Natural transformations, whiskering and the exchange law (Wärn, 28–31), the Yoneda lemma, and full subcategories (Axiom 40 onwards) are not yet treated. It is open whether any of them needs a coherence of level 3.